



Compromised Network Printer | LetsDefend writeup

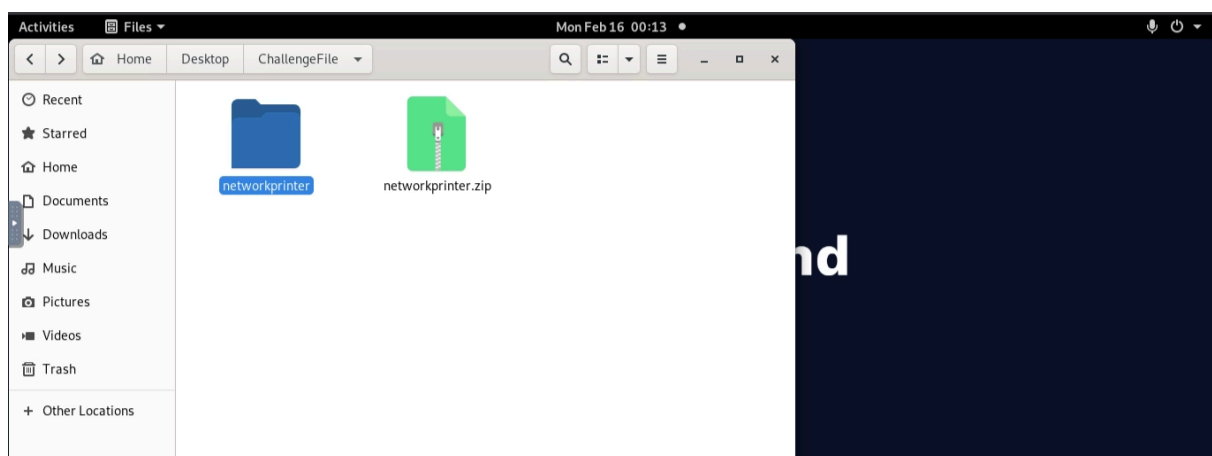
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LetsDefend

Scenario:

You are a DFIR Analyst for a corporation. A network printer running in the internal network has been compromised as it was alerted by our IDS. You have been provided a packet capture to trace the attacker's action and what data was stolen from the printer server.

File location: `/root/Desktop/ChallengeFile/networkprinter.zip`



Question 1:

Identify the port scan activity performed by the attacker on the network. What was the NAT IP Address (Internal IP Address) assigned to the machine being used by the attacker on the internal network?

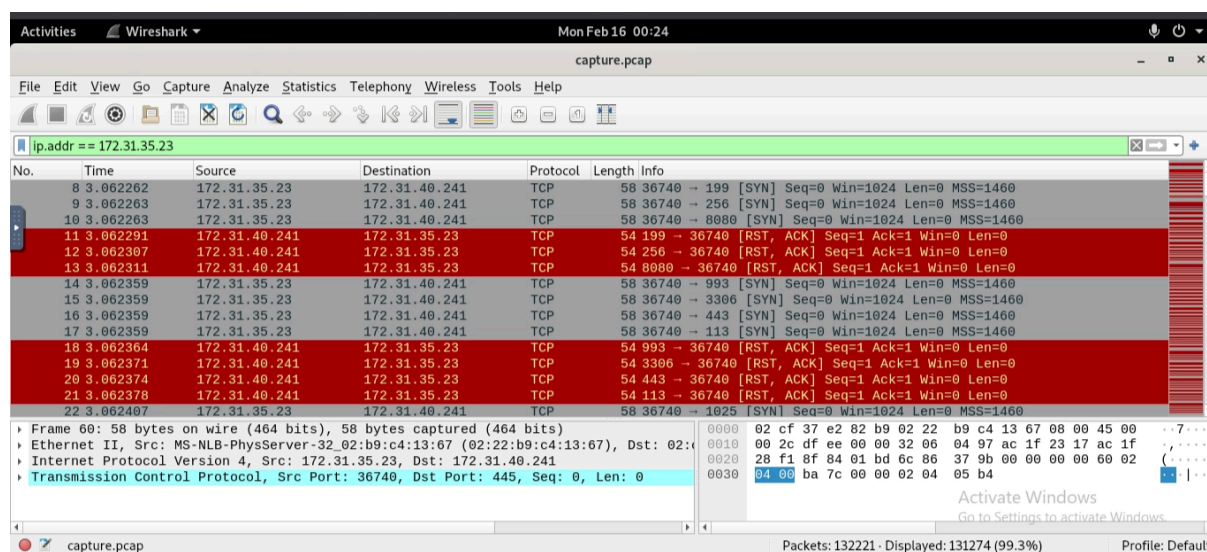
Flag: 172.31.35.23

Process:

As I analyzed the traffic, I noticed this IP `172.31.35.23` sending a bunch of requests, checking different ports.

Known ports being checked:

- 8888
- 443
- 8080
- 80
- etc.
-



The requests were very sequential, which is highly suspicious.

To focus on this, I filtered the traffic for this IP using:

```
ip.addr==172.31.35.23
```

Question 2:

Which ports were open on the network printer? Identify the port used for printer exploitation.

Flag: 9100

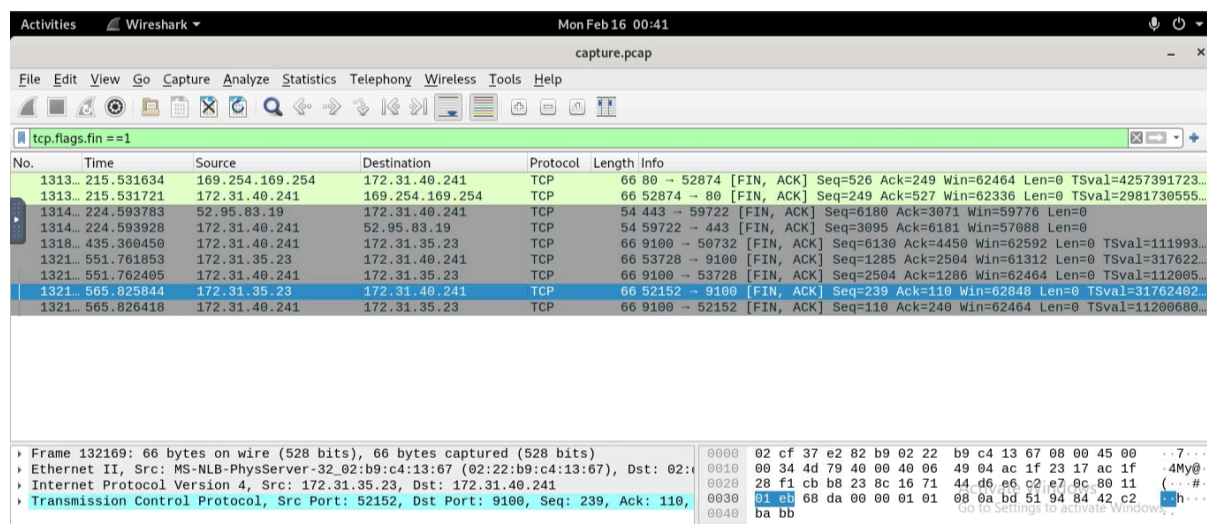
Process:

Let's search for common ports used by a printer.

Sources:

- [CBT Nuggets: Port 9100](#)
- [SonicWall: Network Printer Ports](#)

Next, I filtered TCP responses in Wireshark to see all successfully completed requests (FIN or gracefully finished):



```
tcp.flags.fin==1
```

After locating the target IP **172.31.35.23**, we can see TCP FIN-ACK at port 9100.

Question 3:

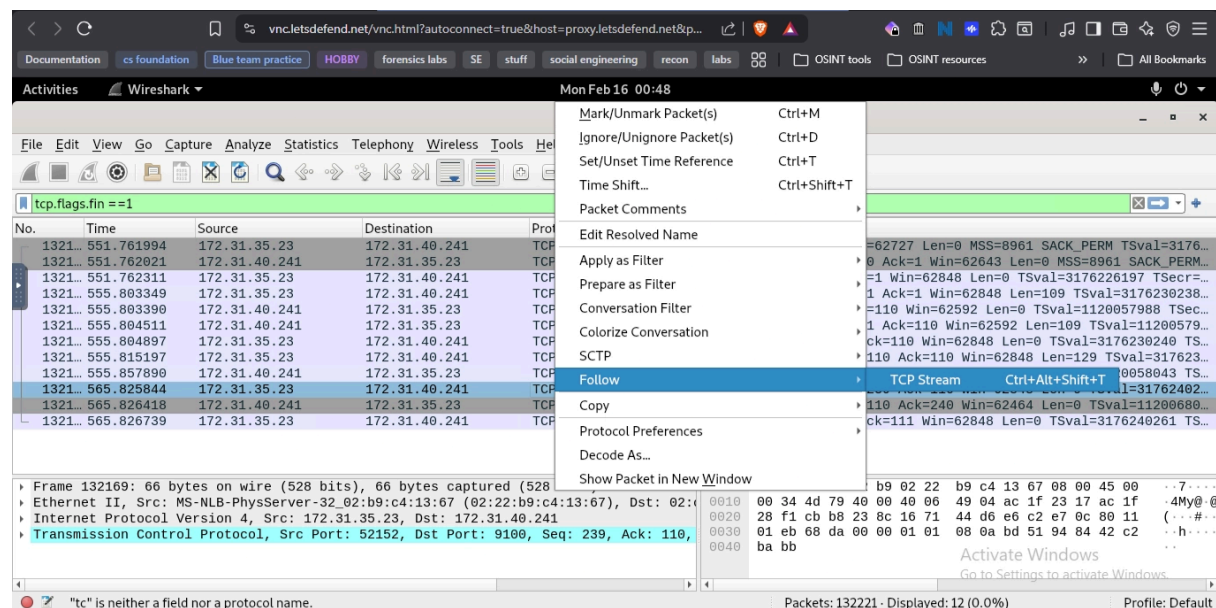
Based on the abused port number, which printer language/method was being abused by the attacker for unattended malicious activity on the network printer?

Flag: PJJ

Process:

What is a printer language/method? Printer languages (also known as Page Description Languages or PDLs) are instructions sent from a computer to a printer, defining how to format, render, and print a document.

Back in Wireshark, I right-clicked the packet we found earlier and selected:



Follow → TCP Stream

Here, we could see the traffic between the two devices.

We found **FSQUERY**. The term **FSQUERY** refers to a File System Query command within the Printer Job Language (PJJ), commonly used with HP and other network-enabled printers to check the file system, retrieve printer status, or manage stored files.

Question 4:

Which Printer Name/Model was attacked?

Flag: HP LaserJet Pro 4001dn

Process:

I filtered the specific port:

```
tcp.port==9100
```

Right-clicked one of the packets and selected:

Follow → TCP Stream

There, we could clearly identify the printer model.

Note: In this section, I mostly relied on my screenshots. Since I hit my free trial limit, I tried to connect the dots manually. Some flags were not included because I couldn't access the full pcap.

Question 5:

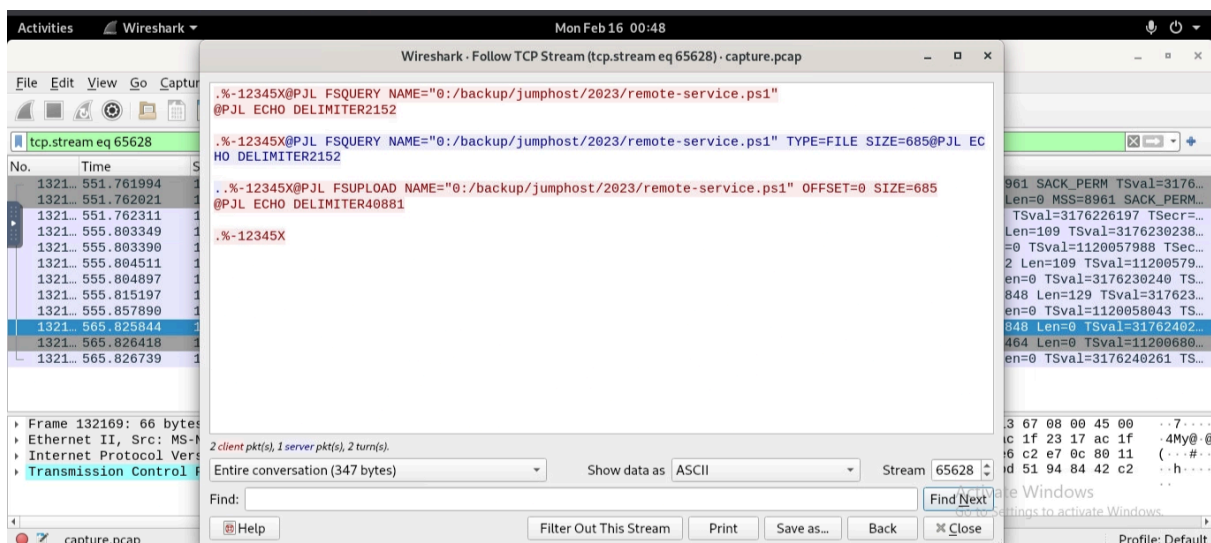
What is the filename of the PowerShell script used by admins which was also found by the attacker?

Flag: remote-service.ps1

Process:

Based on the image we have, we identified `remote-service.ps1` as a PowerShell script.

After searching Google for:



```
fsquery name = some file path printer
```

I found that:

```
FSQUERY NAME="<file-path>"
```

is used in PJJ to check the status or existence of a file on a printer-based disk drive.

Source: [HP Manual](#)

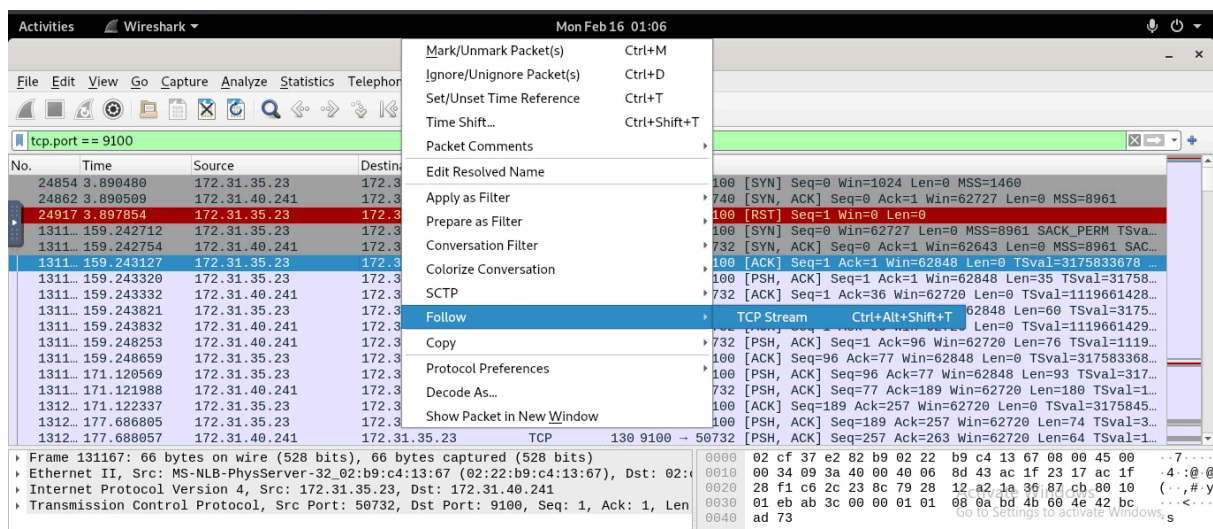
This confirms what we were trying to find. The attacker was querying and retrieving information using PJJ throughout the attack.

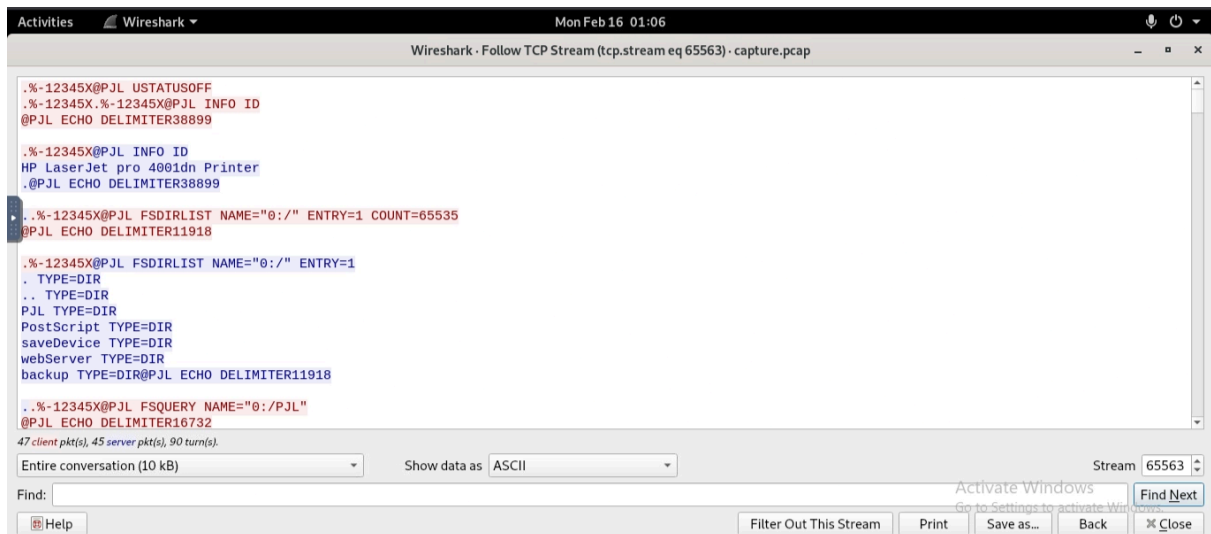
Before this, the attacker executed:

```
FSDIRLIST NAME="0:/"
```

and we could see the directory list, including:

- postscript
- savedevice
- webserver
- `<backup>` → what the attacker found





Question 6:

The attacker found information about RDP within the internal network. What is the directory path where this sensitive information was located?

Flag: /backup/jumphost/2023

Process:

We can assume that the RDP file is located in `jumphost`, as it is associated with `remote-service.ps1`.

This assumption proved correct since there are only a few directories where it could be located, as seen in the attacker's `FSDIRLIST QUERY`.